A project report on

Linear Algebra Playground

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Abstract

Linear algebra is usually taught through symbolic manipulation, which hides the geometry behind it. The Linear Algebra Playground is an interactive web application that makes the core ideas of a first course in linear algebra visible and explorable. It consists of six linked labs covering linear transformations, row reduction and the four fundamental subspaces, span and independence, change of basis, inner products with Gram-Schmidt orthogonalization, and quadratic forms. Every computation is carried out in exact fractions where possible and is recorded as a replayable list of steps, so the output reads like working done by hand. This report describes the objectives, the mathematics implemented, the design of each lab, the computational techniques used, and how the system was tested.

1. Introduction and Objectives

Students can compute a determinant or an RREF correctly and still not understand what the numbers mean. The aim of this project is to close that gap by connecting every algebraic procedure to a picture and to a written explanation.

The project is built around one central idea: a matrix times a vector is a combination of the matrix's columns. From this single statement the rest of the syllabus follows. Row reduction reveals which columns are independent; the pivots give rank, bases and the null space; viewing the matrix as a machine that moves space gives linear transformations and determinants as area scaling; adding a notion of length and angle gives orthogonality; and applying that to a symmetric matrix gives quadratic forms.

The objectives were:

- To cover the syllabus topics (Euclidean space, span, independence, subspaces, basis and dimension, row/column/null spaces, Rank-Nullity, linear transformations, change of basis, inner products, Gram-Schmidt, quadratic forms) in one coherent tool.
- To show the geometry of each concept using draggable 2D and 3D visuals.
- To produce exact, step-by-step working rather than a bare final answer.
- To validate the correctness of the mathematics through automated testing.

2. Mathematical Foundations

This section summarizes the theory that the application implements.

2.1 Vectors, span and independence

A vector in R^n is an ordered list of n real numbers, and the two defining operations are addition and scalar multiplication. A linear combination scales some vectors and adds the results; the span is the set of all such combinations. Vectors $v_1,\ldots,v_k$ are independent if the only solution of $c_1v_1 + \ldots + c_kv_k = 0$ is the trivial one. If a non-trivial solution exists, it is a dependency relation. A basis is an independent spanning set, and the number of vectors in it is the dimension of the space.

2.2 Row reduction and the fundamental subspaces

Gauss-Jordan elimination uses three operations (swap rows, scale a row, add a multiple of one row to another) that preserve the solution set of Ax = 0. The resulting reduced row echelon form (RREF) exposes the pivot columns and free columns. From it the application reads off:

- Column space: the original columns of A at the pivot positions (the set of all outputs Ax).
- Row space: the non-zero rows of the RREF.
- Null space: the solutions of Ax = 0, parametrized by the free variables.

Rank is the number of pivots and nullity is the number of free columns. Since every column is one or the other, the Rank-Nullity Theorem follows: rank + nullity = n, the number of columns. For Ax = b there are three outcomes: no solution (b is outside the column space), exactly one solution (nullity 0), or infinitely many (one particular solution plus any null-space vector).

2.3 Linear transformations and the determinant

A map T is linear if T(u + v) = T(u) + T(v) and T(cu) = cT(u). Every linear map on R^n is multiplication by a matrix, and the columns of that matrix are the images of the standard basis vectors. This is why dragging the tips of the basis arrows in the first lab edits the matrix directly.

The determinant has a geometric meaning: |det A| is the factor by which area (or volume) is scaled, the sign records whether orientation is preserved, and det A = 0 means space collapses onto a lower dimension. For an n×n matrix,

the following statements are equivalent: det A ≠ 0, A is invertible, rank = n, nullity = 0, the columns are independent, the columns form a basis of R^n, and Ax = b has a unique solution for every b.

2.3.1 Eigenvalues

An eigenvector satisfies Av = λv, so A only stretches or flips it. Eigenvalues are the roots of det(A − λI) = 0. For a 2×2 matrix, λ = (trace ± √(trace² − 4·det)) / 2. A rotation has complex eigenvalues, so it has no real fixed direction.

2.4 Coordinates and change of basis

Coordinates are an address relative to a chosen basis. If B has the basis vectors as columns, then $[v]_B = B^{-1}v$. Converting between two bases uses the transition matrix $P = B'^{-1}B$. The matrix of a transformation in basis B is $[T]_B = B^{-1}AB$. Such matrices are called similar; they describe the same transformation and share determinant, trace and eigenvalues.

2.5 Inner products and orthogonality

The dot product gives length, ‖u‖ = √(u·u), and angle, cos θ = (u·v)/(‖u‖‖v‖). The projection of v onto u is ((u·v)/(u·u))·u. An inner product generalizes the dot product to any symmetric, linear, positive-definite rule, for example the weighted form $\langle u,v\rangle = u^TWv$. Its matrix representation is the Gram matrix, $G[i][j] = v_i \cdot v_j$.

Gram-Schmidt converts independent vectors into an orthonormal set spanning the same space: subtract from each vector its projections onto the earlier ones, then divide by the length. Placing the result as columns of Q gives $Q^TQ = I$, and writing A = QR gives the QR decomposition with $R = Q^TA$ upper triangular.

2.6 Quadratic forms

For a symmetric matrix S, the quadratic form is $q(x) = x^TSx$. It is classified as positive definite (bowl), negative definite (inverted bowl), indefinite (saddle) or semidefinite (trough). Two equivalent tests are used: the signs of the eigenvalues, and Sylvester's criterion on the leading principal minors. A non-symmetric matrix is replaced by its symmetric part $(A + A^T)/2$, since x^TAx depends only on that part.

3. System Design and Features

The application is organized as six labs plus a shared computation core. The 2D labs are drawn on an HTML Canvas, and the 3D labs use three.js. All labs share

the same exact-arithmetic engine, so a result seen in one lab is consistent with every other.

	Lab Topic	What the user can do
1	Linear transformations and determinant	Drag the arrows î and ĵ; the matrix updates live. The unit square is shaded blue for positive determinant and orange for negative. Collapse (det = 0) is shown visibly. An animation slider morphs the plane from the identity to A.
2	Matrix lab: RREF and subspaces	Enter matrices up to 4×5; step through row reduction with Prev/Next/Play; see pivot and free columns, bases for the column space (teal), row space (purple) and null space (orange), the Rank-Nullity banner, and 3D planes for 3×3 cases. A subspace tester checks the three subspace rules and shows a counterexample when one fails.
3	Span and independence	Use sliders $c_1 \ldots c_4$ to build a linear combination with tip-to-tail arrows; the span is shaded. The tool reports independence, the dependency relation, a basis and the dimension, and tests whether a target b lies in the span.
4	Change of basis	Two coordinate grids (basis B in blue, B′ in green) are drawn over the standard grid. The vector is decomposed along each, the transition matrix is shown, and a commutative-square diagram verifies [T]_B. Inputs that do not form a basis are rejected with the reason.
5	Inner products and Gram-Schmidt	Part A computes dot product, length, angle and projection with the right-angle marker. Further parts support weighted inner products (validated for positive definiteness), the Gram matrix, and an animated Gram-Schmidt that ends by displaying $Q^TQ = I$ and the QR factors.

6	Quadratic forms	Edit a symmetric matrix and watch the surface z = q(x,y) change between bowl, hill and saddle. Eigenvalue and Sylvester tests are both run and cross-checked, with principal axes drawn on the contour plot.

A separate Studio view displays several properties of one matrix at once: determinant, inverse, rank, eigenvalues and real eigenvector directions (amber lines).

Design principle: explain, don't just answer. Each algorithm returns not only a result but a list of steps with a matrix snapshot at each stage. The steps panel replays these like a teacher working at a board, which is what separates the tool from a plain calculator.

4. Computational Methods

Several numerical and algorithmic decisions were made to keep the output both correct and readable.

Exact fractions. Computers store decimals approximately (0.1 + 0.2 gives 0.30000000000000004). When inputs are integers or simple decimals, values are stored as fractions with big-integer numerators and denominators, so 1/3 stays exactly 1/3. A badge indicates whether Exact or Float mode is active.

Tolerance. In decimal mode, any value smaller than 10^{-10} in magnitude is treated as zero, preventing rounding noise from making a singular matrix appear invertible.

Partial pivoting. With decimals, the largest entry in the column is chosen as pivot to avoid dividing by a tiny number and amplifying error.

Determinant and inverse. Determinants use cofactor expansion for 2×2 and 3×3 matrices so that the working can be displayed, and row reduction for 4×4. The inverse is found by Gauss-Jordan on the augmented matrix [A | I]; if a zero row appears in the left half, the matrix is singular.

Eigenvalues. For symmetric 3×3 matrices the Jacobi method applies small rotations repeatedly to eliminate off-diagonal entries until only the eigenvalues remain on the diagonal. For 2×2 matrices the closed formula is used.

Safe angle computation. Rounding can push a cosine slightly outside [−1, 1] (for example 1.0000000002), which would make the inverse cosine fail, so the value is clamped before use.

Gram-Schmidt edge cases. If some u_k is the zero vector, then v_k was dependent on earlier vectors, and the tool skips it and explains why. The output also depends on the input order, which the tool notes.

5. Testing and Verification

Because the tool is meant to teach, a wrong answer is worse than no answer, so correctness was treated as a first-class requirement. The test suite checks results against known hand-calculated examples, such as the RREF, null space and Gram-Schmidt worked examples, and checks mathematical identities on many generated inputs. In particular, the Rank-Nullity Theorem (rank + nullity = n) is verified on hundreds of random matrices. Other identities that can be checked automatically include $A \cdot A^{-1} = I$ for invertible matrices, $Q^T Q = I$ for Gram-Schmidt output, $A \cdot x = 0$ for every null-space basis vector, and agreement between the eigenvalue test and Sylvester's criterion for quadratic forms.

Edge cases were tested deliberately: singular matrices, dependent input vectors, vectors that do not form a basis, non-symmetric matrices passed to the quadratic-form lab, and invalid weight matrices for the inner product. In each case the tool gives a clear message explaining the reason instead of failing silently.

6. Results and Observations

The finished application demonstrates the full syllabus in one place and satisfies the stated objectives.

- **Coverage:** every syllabus topic is mapped to a specific lab and visual, from Euclidean space (all canvases) to quadratic forms (Lab 6).
- **Connection of algebra and geometry:** dragging the basis arrows shows immediately that the columns of a matrix are where the basis vectors land; shrinking a determinant to zero shows the plane collapsing and the inverse ceasing to exist.
- **Unity of concepts:** the Studio readout shows determinant, rank, nullity and invertibility moving together, illustrating the equivalence of the invertibility statements in Section 2.3.

- **Transparency:** exact arithmetic and replayable steps let a student check the tool's working against their own notebook.
- **A known artifact, explained:** linearly blending the identity with a rotation, $M(t) = (1 - t)I + tA$, makes a shape shrink mid-animation (for a 90° rotation at $t = 0.5$, $\det M = 0.5$). This is a property of linear blending, not a bug, and a "rotation-true" option is provided to avoid it.

7. Limitations and Future Work

Current limitations.

- The matrix lab accepts up to 4×5 matrices; vectors in R^4 and R^5 are computed but not drawn.
- The Jacobi eigenvalue routine is specialized to symmetric 3×3 matrices, and general non-symmetric matrices of larger size are not handled for eigen-analysis.
- Exact fractions can grow large for long chains of operations, and decimal inputs fall back to floating-point mode with tolerance.

Possible extensions.

- Add singular value decomposition (SVD) and diagonalization labs, completing the path from eigenvalues to data applications.
- Add least-squares fitting, which follows naturally from projection.
- Support larger matrices and general eigenvalue computation (for example, QR iteration).
- Allow saving and sharing a lab state through a link, and add guided exercises with automatic checking.

8. Conclusion

The Linear Algebra Playground turns a first course in linear algebra into something that can be seen, manipulated and checked. By building every topic from the single idea that a matrix times a vector is a combination of its columns, the project shows linear algebra as one connected subject rather than a list of separate procedures. Exact arithmetic, step-by-step explanations and automated testing make it reliable as well as illustrative. The work deepened understanding of the theory (subspaces, Rank-Nullity, change of basis, orthogonality and definiteness) and of the practical issues in implementing it, such as numerical stability and tolerance, and it provides a solid base for further extensions.

References

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